Load Adjust Signal Malfunctions

Symptoms

- Load adjust signal **not** available from customer interface box.

How To Use This Tree

This symptom tree can be used to troubleshoot engine symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check Customer Interface Box Wiring	
	STEP 1A. Check Droop Adjust Potentiometer Supply Wire	Less than 10 ohms resistance?
	STEP 1B. Check Droop Adjust Potentiometer Return Wire	Less than 10 ohms resistance?
	STEP 1C. Check Frequency Adjust Signal Wire	Less than 10 ohms resistance?
STEP 2.	Check Engine Harness to Customer Interface Box Cable	
	STEP 2A. Check Frequency Adjust Supply and Signal Wires	Less than 10 ohms resistance?
	STEP 2B. Check Frequency Adjust Return and Signal Wires	Less than 10 ohms resistance?

STEP 1. Check Customer Interface Box Wiring

STEP 1A. Check Droop Adjust Potentiometer Supply Wire

Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.

Action	Specification/Repair	Next Step
Check the droop adjust potentiometer supply wire. • Place one test lead on the droop adjust potentiometer 5 volt supply (sensor supply 1) pin in connector C3. • Place the other test lead on the droop adjust potentiometer 5 volt supply (sensor supply 1) terminal on the X4 connector.	Less than 10 ohms resistance? **YES**	**1B**
	Less than 10 ohms resistance? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	**Repair complete.**

STEP 1B. Check Droop Adjust Potentiometer Return Wire

Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.

Action	Specification/Repair	Next Step

Check the droop adjust potentiometer return wire. • Place one test lead on the droop adjust potentiometer return (sensor return 1) pin in connector C3. • Place the other test on droop adjust potentiometer return (sensor return 1) terminal on the X4 connector.	Less than 10 ohms resistance? **YES**	**1C**
	Less than 10 ohms resistance? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	**Repair complete.**

STEP 1C. Check Frequency Adjust Signal Wire

Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.

Action	Specification/Repair	Next Step
Check the frequency adjust signal wire. • Place one test lead on the generator output frequency adjust potentiometer signal pin in connector C3. • Place the other test lead on the generator output frequency adjust potentiometer signal terminal on the X4 connector.	Less than 10 ohms resistance? **YES**	**2A**
	Less than 10 ohms resistance? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	**Repair complete.**

STEP 2. Check Engine Harness to Customer Interface Box Cable

STEP 2A. Check Frequency Adjust Supply and Signal Wires

Conditions :

- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box
- Disconnect customer interface box to engine harness cable connector C9 from the engine harness.

Action	Specification/Repair	Next Step
Check frequency adjust supply and signal wires. • Place a jumper between the droop adjust potentiometer 5 volt supply (sensor supply 1) pin and the generator output frequency adjust potentiometer signal pin in the C9 connector. • Place one test lead in the droop adjust potentiometer 5 volt supply (sensor supply 1) pin of the C3 connector. • Place the other test lead in the generator output frequency adjust potentiometer signal pin of the C3 connector.	Less than 10 ohms resistance? **YES**	**2B**
	Less than 10 ohms resistance? **NO** **Repair :** Replace the cable.	**Repair complete.**

STEP 2B. Check Frequency Adjust Return and Signal Wires

Conditions :

- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.
- Disconnect customer interface box to engine harness cable connector C9 from the engine harness.

Action	Specification/Repair	Next Step
Check frequency adjust return and signal wires. • Place a jumper between the generator output frequency adjust potentiometer signal pin and the droop adjust potentiometer return (sensor return 1) pin in the C9 connector. • Place one test lead in the droop adjust potentiometer return (sensor return 1) pin of the C3 connector. • Place the other test lead in the generator output frequency adjust potentiometer signal pin of the C3 connector.	Less than 10 ohms resistance? **YES** **Repair :** Refer to Section TF in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850, Modular Common Rail System, Series Engines, Bulletin 4021493 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021493.html) or refer to the OEM Service Manual for potentiometer repair instructions.	**Repair complete.**
	Less than 10 ohms resistance? **NO** **Repair :** Replace the cable.	**Repair complete.**

Last Modified: 09-Aug-2006